



(Link: www.nhs.uk/)

Causes

Parkinson's disease

— Overview (Link: www.nhs.uk/conditions/parkinsons-disease/)

- —Symptoms (Link: www.nhs.uk/conditions/parkinsons-disease/symptoms/)
- —**Causes**
- —Diagnosis (Link: www.nhs.uk/conditions/parkinsons-disease/diagnosis/)
- —Treatment (Link: www.nhs.uk/conditions/parkinsons-disease/treatment/)
- —Living with (Link: www.nhs.uk/conditions/parkinsons-disease/living-with/)

Parkinson's disease is caused by a loss of nerve cells in the part of the brain called the substantia nigra.

Nerve cells in this part of the brain are responsible for producing a chemical called dopamine.

Dopamine acts as a messenger between the parts of the brain and nervous system that help control and co-ordinate body movements.

If these nerve cells die or become damaged, the amount of dopamine in the brain is reduced.

This means the part of the brain controlling movement cannot work as well as normal, causing movements to become slow and abnormal.

The loss of nerve cells is a slow process. The symptoms of Parkinson's disease usually only start to develop when around 50% of the nerve cell activity in the substantia nigra have been lost.

What causes the loss of nerve cells?

It's not known why the loss of nerve cells associated with Parkinson's disease occurs, although research is ongoing to identify potential causes.

Currently, it's believed a combination of genetic changes and environmental factors may be responsible for the condition.

Genetics

A number of genetic factors have been shown to increase a person's risk of developing Parkinson's disease, although exactly how these make some people more susceptible to the condition is unclear.

Parkinson's disease can run in families as a result of faulty genes being passed to a child by their parents. But it's rare for the disease to be inherited this way.

Environmental factors

Some researchers also feel environmental factors may increase a person's risk of developing Parkinson's disease.

It's been suggested that pesticides and herbicides used in farming and traffic or industrial pollution may contribute to the condition.

But the evidence linking environmental factors to Parkinson's disease is inconclusive.

Other causes of parkinsonism

"Parkinsonism" is the umbrella term used to describe the symptoms of tremors, muscle rigidity and slowness of movement.

Parkinson's disease is the most common type of parkinsonism, but there are also some rarer types where a specific cause can be identified.

These include parkinsonism caused by:

- medication (drug-induced parkinsonism) – where symptoms develop after taking certain medications, such as some types of antipsychotic medication, and usually improve once the medication is stopped
- other progressive brain conditions – such as progressive supranuclear palsy (Link: www.nhs.uk/conditions/progressive-supranuclear-palsy-psz/), multiple system atrophy and corticobasal degeneration (Link: www.nhs.uk/conditions/corticobasal-degeneration/)
- cerebrovascular disease – where a series of small strokes (Link: www.nhs.uk/conditions/stroke/) cause several parts of the brain to die

You can read more about parkinsonism (Link: <http://www.parkinsons.org.uk/content/types-parkinsons-and-parkinsonism>) on the Parkinson's UK website.

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